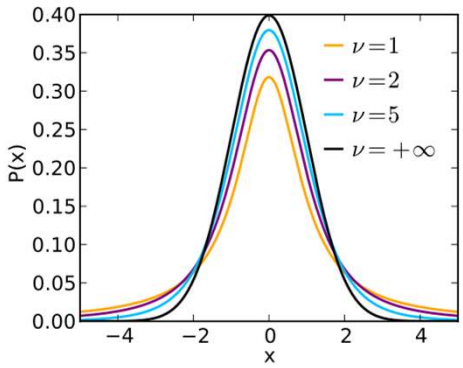
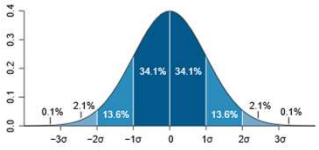
Agenda

- Last time
  - Informal approach to confidence intervals
  - Approximate distribution of the sample mean using CLT
  - Approximate distribution of the sample proportion using CLT
- Today
  - The “student” distribution



1

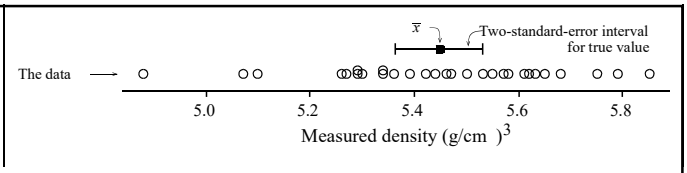
Logic of sampling



- The distance between $\bar{x}$ and μ is hardly bigger than $2\text{sd}(\bar{X})$ if we have a normal distribution for $\bar{X}$
  1. This could have been practically useless because nobody knows that their data is coming from a normal distribution
  2. No one knows the value of $\text{sd}(\bar{X})$
- However, CLT rescues us saying, yes,
  - The population is normal in many cases, i.e., $\bar{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Further, we have an even more convenient result $\bar{X} \approx \mathcal{N}(\mu, \text{se}(\bar{x}))$

2

Example



- In 1798 Cavendish measured the density of the earth
  - sample mean $\bar{x}$ = 5.447931 g/cm^3 and s_x = 0.2209457 g/cm^3
  - Therefore se($\bar{x}$) = 0.04102858
- From the theory, the true mean is $\bar{x} = 5.447931 \pm 2 \times 0.04102858$
- The number of digits of precision is unwarranted so we can say that Cavendish predicted true value between 5.37 and 5.53
- Current accepted value is ~ 5.517

5.50	5.61	4.88	5.07	5.26	5.55	5.36	5.29	5.58	5.65
5.57	5.53	5.62	5.29	5.44	5.34	5.79	5.10	5.27	5.39
5.42	5.47	5.63	5.34	5.46	5.30	5.75	5.68	5.85	

3

Review

- Why is the standard deviation of $\bar{X}$ not a useful measure of the precision of $\bar{X}$ as an estimator in practical applications?
- What measure of precision do we use in practice?
- How is se($\bar{x}$) related to the standard deviation of $\bar{X}$?
- When we use the formula se$(\bar{x}) = s_x/\sqrt{n}$, what is s_x and how do you obtain it?

4

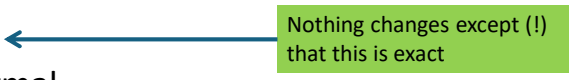
Beyond Approximation

- Although a good practical tool, there are two approximations
  - The CLT approximation for large n
  - The "se($\bar{x}$)" approximation for the unknown standard deviation
- Both approximations are problematic for small n
- If the population were known to be exactly normal, then we have exact results

5

Distribution of Sample mean

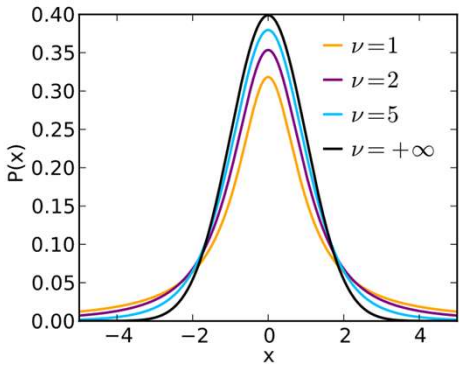
- For independent identical (iid) **normal** random variables, and $Y = X_1 + X_2 + \cdots X_n$ with $\bar{X} = Y/n$, $\mathbb{E}[X_i] = \mu$ and finite variance σ^2
  - $\mathbb{E}[\bar{X}] = \mu, \mathrm{Var}(\bar{X}) = \sigma^2/n$
  - $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
  - $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ is standard normal
- $\bar{x}$ is an estimate of the sample mean $\bar{X}$
  - $\bar{x} = \sum_i x_i \,/n$
  - This formula comes from the estimator $\bar{X}$ by simply observing the values



6

The student t-distribution

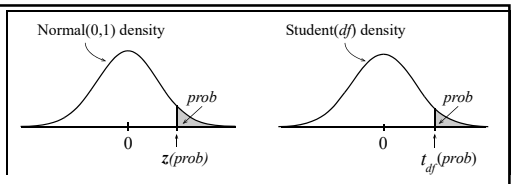
- If the population were known to be exactly normal, then we have the result $T_n = (\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - This is the student t-distribution with parameter "n-1"
- Here we have introduced
  - A new statistic S (it is a statistic not an estimator)
  - A new distribution T_n (df) where "df" stands for degrees of freedom



This is new, we get to use "S" instead of σ

7

Statistic for the variance



- S is easily recognizable
  - We define the random variable $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$
  - S is the square root
- Thus, we have the practical result $(\bar{X} - \mu)/\mathrm{se}(\bar{X}) \sim t_{n-1}$
- For small n,
  - using t-values rather than z-values is robust even for non-normal distribution (empirical result)
  - For proportions, when the binomial assumption is valid, t is not needed
- For large n, t-distribution tends to the standard normal anyway
- In stat courses, tables for the t-distribution are available

8

Extracts from the Student's t-Distribution Table

prob — Desired *upper-tail prob*

Desired *df* →

df	.20	.15	.10	.05	.025	.01	.005	.001	.0005	.0001
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
…	…	…	…	…	…	…	…	…	…	…
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
…	…	…	…	…	…	…	…	…	…	…
15	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
…	…	…	…	…	…	…	…	…	…	…
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291	3.719

Some results

… and we just introduced T out of the blue!

9

#1: Expectation of S^2

- We can show that $E(S^2) = \sigma^2$
  - Given $S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$ we can show that $S^2 = \frac{\sum_{i=1}^{n} X_i^2 - n\overline{X}^2}{n-1}$
  - $E[(n-1)S^2] = E\left(\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right) = nE(X_i^2) - nE(\overline{X}^2)$
    - But $E(W^2) = \mathrm{Var}(W) + E(W)^2$
  - $E[(n-1)S^2] = n\,\mathrm{Var}(X_1) + nE(X_1)^2 - n\,\mathrm{Var}(\overline{X}) - nE(\overline{X})^2$
  - $E[(n-1)S^2] = n\sigma^2 + n\mu^2 - n(\sigma^2/n) - n\mu^2 = (n-1)\sigma^2$
- Note: this is not the expectation of S

10

$$\sum_{i=1}^{N}(X_i - \bar{X})^2 = \sum_{i=1}^{N}\left[(X_i - \mu) - (\bar{X} - \mu)\right]^2$$

$$= \sum_{i=1}^{N}(X_i - \mu)^2 - 2(\bar{X} - \mu)\sum_{i=1}^{N}(X_i - \mu) + \sum_{i=1}^{N}(\bar{X} - \mu)^2.$$

But

$$\sum_{i=1}^{N}(X_i - \mu) = \sum_{i=1}^{N} X_i - N\mu = N(\bar{X} - \mu),$$

and

$$\sum_{i=1}^{N}(\bar{X} - \mu)^2 = N(\bar{X} - \mu)^2.$$

Therefore

$$\boxed{\sum_{i=1}^{N}(X_i - \bar{X})^2 = \sum_{i=1}^{N}(X_i - \mu)^2 - N(\bar{X} - \mu)^2}$$

11

Bessel Correction

- If we had started with n in the denominator
- Then we have ended up with $nE(S^2) = (n-1)\,\sigma^2$
- In such the expectation would have a (n-1)/n factor
  - This is the Bessel's correction
  - For n = 2, we would have ended up with E(S^2) = ½ σ^2
  - This is exactly what we saw in the experiment on the first day of class

12

Experiment

Now focusing on the last two columns

All possible samples of size 2	$\bar{b} = \frac{\sum x_i}{n}$	$\frac{\sum(x-\bar{b})^2}{n-1}$	$\frac{\sum(x-\bar{b})^2}{n}$
0 0	(0+0)/2 = 0	$((0-0)^2+(0-0)^2)/1 = 0$	0
0 2	1	$((0-1)^2+(2-1)^2)/1 = 2$	1
0 4	2	8	4
2 0	1	2	1
2 2	2	0	0
2 4	3	2	1
4 0	2	8	4
4 2	3	2	1
4 4	4	0	0

- Considering samples of size two,
  - Dividing by n results in $E[\overline{s}^2] = \frac{12}{9} = \frac{4}{3}$
  - Division by n underestimates the true variance
  - Dividing by (n-1) results in $E[\overline{s}^2] = \frac{24}{9} = \frac{8}{3}$

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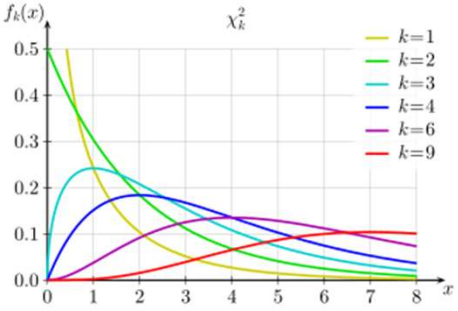
#2: Independence

- For independent identical (iid) **normal** random variables X_i with mean μ and variance σ^2
  - $\bar{X}$ is normal with mean μ and $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$
- Both $\bar{X}$ and S^2 are computed from the same data, yet they are independent, this is a surprising result (not proven here)
- Further
  - $\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma}\right)^2$ which is the sum of standard normals from *iid* X_i
  - The rhs has a name! $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$
  - Thus distribution of the statistic $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

15

Chi-square distribution [1/3]

- If $Z_1, Z_2, \ldots, Z_n$ are independent *standard* normal then
  - $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is said to have a chi-square distribution with n degrees of freedom and is denoted as $X \sim \chi_n^2$
  - $f_X(x) = \frac{x^{n/2-1}e^{-x/2}}{2^{n/2}\Gamma(n/2)}$
  - $\Gamma(y) = \int_0^\infty x^{y-1}\, e^{-x} dx$
- Example (n=1)
  - $Z_1 \sim N(0,1), X = Z_1^2$
  - $f_X(x) = \frac{e^{-x/2}}{\sqrt{x}\sqrt{2\pi}}$



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Chi-square distribution [2/3]

- In stat courses, tables for the chi-square distribution are available for different number of degrees of freedom, and for different values of the probability
- MGF: $\Phi_X(t) = (1-2t)^{-n/2}$ (t < ½)
- Additive property
  - If X1 and X2 are *independent* chi-square random variables with n1 and n2 degrees of freedom respectively, then X1 + X2 is also a chi-square random variable with n1 + n2 degrees of freedom

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Chi-square distribution [3/3]

- Can we use the additive property?
  - $(n-1)S^2 = \sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 = \sum_{i=1}^{n}(X_i-\mu)^2 - n\left(\overline{X}-\mu\right)^2$
  - $\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} = \frac{\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2}{\sigma^2} + \frac{n\left(\overline{X}-\mu\right)^2}{\sigma^2}$
  - $\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2}{\sigma^2} + \left(\frac{\sqrt{n}\left(\overline{X}-\mu\right)}{\sigma}\right)^2$

Recognize standard normal

Recognize as χ_n^2

Recognize as $(n-1)/\sigma^2 S^2$

Independent